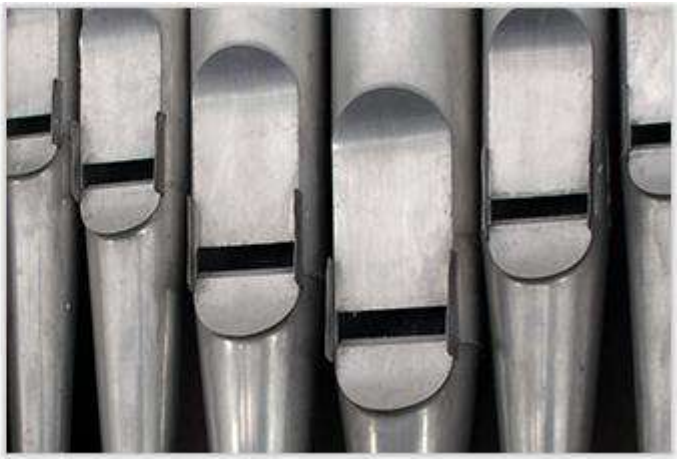
Description

Image





Caption

1. Organ pipes are made of tin or of a tin - lead alloy. 2. Close-up of the material. © Håkan Svensson (Xauxa) at en.wikipedia - (CC BY-SA 3.0)

The material

Tin (symbol Sn) has been known to mankind since at least 3500 BC. The discovery that copper alloyed with tin to give bronze, greatly improving the mechanical properties, launched the Bronze age. In 1800 Napoleon offered prize of 12,000 francs for a method for preserving food for his armies. The tin can (steel coated with tin), which revolutionize the storage and preservation of foodstuffs and liquids, was invented in 1810, ironically by an Englishman; the first commercial canning factory opened just 3 years later.

Compositional summary

Tin, Sn.

General properties

Density		7.26e3	-	7.27e3	kg/m^3
Price	*	19.4	-	23.1	GBP/kg
Date first used		-3500			

Mechanical properties

Young's modulus	41	-	45	GPa
Shear modulus	14	-	18	GPa
Bulk modulus	38	-	46	GPa
Poisson's ratio	0.325	-	0.335	
Yield strength (elastic limit)	7	-	15	MPa
Tensile strength	11	-	18	MPa
Compressive strength	7	-	15	MPa

Elongation		55	-	75	% strain
Hardness - Vickers		3	-	5	HV
Fatigue strength at 10^7 cycles	*	4	-	9	MPa
Fracture toughness	*	15	-	30	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.015	-	0.045	

Thermal properties

Melting point		230	-	232	°C
Maximum service temperature	*	90	-	100	°C
Minimum service temperature		0	-	13.2	°C
Thermal conductor or insulator?		Good conductor			
Thermal conductivity		60	-	61.5	W/m.°C
Specific heat capacity		216	-	228	J/kg.°C
Thermal expansion coefficient		22.5	-	23.5	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good conductor			
Electrical resistivity		10	-	12	µohm.cm

Optical properties

Transparency	Opaque

Critical Materials Risk

High critical material risk?	Yes

Processability

Castability	5			
Formability	4	-	5	
Machinability	5			
Weldability	1			
Solder/brazability	5			

Durability: water and aqueous solutions

Water (fresh)	Excellent
Water (salt)	Acceptable
Soils, acidic (peat)	Excellent
Soils, alkaline (clay)	Excellent
Wine	Acceptable

Durability: acids

Acetic acid (10%)	Excellent
Acetic acid (glacial)	Excellent
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Acceptable
Hydrochloric acid (36%)	Limited use
Hydrofluoric acid (40%)	Limited use
Nitric acid (10%)	Unacceptable
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Acceptable
Phosphoric acid (85%)	Limited use
Sulfuric acid (10%)	Unacceptable
Sulfuric acid (70%)	Unacceptable

Durability: alkalis

Sodium hydroxide (10%)	Limited use
Sodium hydroxide (60%)	Limited use

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Excellent
Crude oil	Limited use
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Acceptable
Acetone	Excellent
Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	Excellent
Formaldehyde (40%)	Excellent

Glycerol	Excellent
Methyl alcohol (methanol)	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Unacceptable
Fluorine (gas)	Unacceptable
O2 (oxygen gas)	Limited use
Sulfur dioxide (gas)	Excellent

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Acceptable
Tolerance up to 250 C (482 F)	Unacceptable
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	2.68e5	tonne/yr
Reserves, principal component	3.94e6	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 11.9	-	13.1	kg/kg
Embodied energy (virgin grade production)	* 165	-	182	MJ/kg
Water usage	* 1.04e4	-	1.15e4	l/kg

Material processing: energy

Metal casting, embodied energy	* 31.4	-	35	MJ/kg
Metal extrusion, hot, embodied energy	* 1.25	-	1.84	MJ/kg
Metal rolling and forging, embodied energy	* 1.06	-	1.41	MJ/kg
Wire drawing, embodied energy	* 13.1	-	14.5	MJ/kg

Metal powder forming, embodied energy	* 7.26	-	8.4	MJ/kg
Chemical vapor deposition (CVD), embodied energy	* 2.04e3	-	2.66e3	MJ/kg
Machining, coarse, embodied energy	* 14.7	-	16.2	MJ/kg
Machining, fine, embodied energy	* 14.8	-	16.5	MJ/kg
Machining, grinding, embodied energy	* 14.9	-	16.8	MJ/kg
Machining, laser, CO2, embodied energy	* 0.243	-	0.317	MJ/m

Material processing: climate change

Metal casting, climate change (CO2-eq)	* 2.09	-	2.34	kg/kg
Metal extrusion, hot, climate change (CO2-eq)	* 0.0817	-	0.121	kg/kg
Metal rolling and forging, climate change (CO2-eq)	* 0.0696	-	0.0923	kg/kg
Wire drawing, climate change (CO2-eq)	* 0.86	-	0.95	kg/kg
Metal powder forming, climate change (CO2-eq)	* 0.548	-	0.637	kg/kg
Chemical vapor deposition (CVD), climate change (CO2-eq)	* 168	-	219	kg/kg
Machining, coarse, climate change (CO2-eq)	* 0.959	-	1.06	kg/kg
Machining, fine, climate change (CO2-eq)	* 0.966	-	1.08	kg/kg
Machining, grinding, climate change (CO2-eq)	* 0.973	-	1.1	kg/kg
Machining, laser, CO2, climate change (CO2-eq)	* 0.02	-	0.0261	kg/m

Recycling and end of life

Recycle	✓			
Functional recycle	✓			
Climate change (CO2-eq), recycling	* 2.38	-	2.62	kg/kg
Embodied energy, recycling	* 36.3	-	40	MJ/kg
Recycle fraction in current supply	22			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Tin(II) salts can be poisonous by ingestion and other routes, and there is evidence that tin can have experimental carcinogenic and human mutagenic effects. Some organo-tin compounds are very toxic.

Supporting information

Technical notes

Tin is extracted by the reduction of cassiterite, SnO2, with carbon. At normal temperatures tin is metallic ("white" tin), but below 13.2 C it transforms (slowly) to non-metallic gray tin -- a problem known as "tin pest" when tin is used at low temperatures.

Typical uses

Tin is used in pure form in storage tanks for pharmaceutical chemical solutions, as electrodes of capacitors, and fuse wire and as organ pipes (though usually alloyed with some lead). Its most important applications, however, are as a coating on steel sheet ("tin plate") and as an alloying element in bronze, pewter and solder. Its salts are used as polymer additives, for antifouling paints, and to produce a transparent, conducting coating on glass.

Further reading

Eco data from Hammond, G. and Jones, C. (2006) "Inventory of carbon and energy (ICE), Dept. of Mechanical Engineering, University of Bath, UK

Links

ProcessUniverse

Producers

Reference